

DENTORACULUS - A gamified Virtual Reality experience for dental morphology learning

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ABSTRACT

Dentoraculus is an educational tool on VR for the teaching of dental morphology whose objective is to train dental students in a didactic and playful way, as well as to complement the teaching of complex theoretical concepts and laboratory training. To enhance the motivation, gamification and storytelling techniques were applied. 3D models of dental pieces were obtained using the photogrammetry technique. Finally, experience was tested with teachers and students, obtaining satisfactory result.

Index Terms: Dentistry—Gamification—Virtual Reality—Storytelling—Photogrammetry;

1 INTRODUCTION

The subject of Dental Morphology is part of the curriculum of the Dentistry course taught at the undergraduate level of higher education and consists of the morphological and functional description of the types of teeth (incisors, canines, premolars, molars).

One of the main challenges in teaching dental morphology is that the acquisition of theoretical knowledge requires spatial imagination [2]. Traditional tools such as mannequins or mock-ups have limitations in the quantity and quality of feedback [1]. We believe that VR can become an educational tool that not only overcomes these limitations, but also encourages active participation and increases student motivation and engagement.

Consequently, with the collaboration of dental professors, a laboratory experience for teaching dental morphology based on Virtual Reality was developed to allow students to practice as often as required and to learn at their own pace.

With the aim of increasing the motivation and active participation of the students, we emphasized in the design of the experience making it as interactive and playful as possible, as well as involving the student as part of a story where he/she is the protagonist and generating an emotional connection.

2 METHODS AND MATERIALS

Dentoraculus experience was developed in six stages: content planning, storytelling, design and sculpting of 3D models and environments, programming, internal laboratory testing and user testing with the participation of teachers and students as shown in Figure 2.

The following paragraphs describe characteristics of the virtual environment, which provides technical details about software and devices; design of the VR experience that explains the learning

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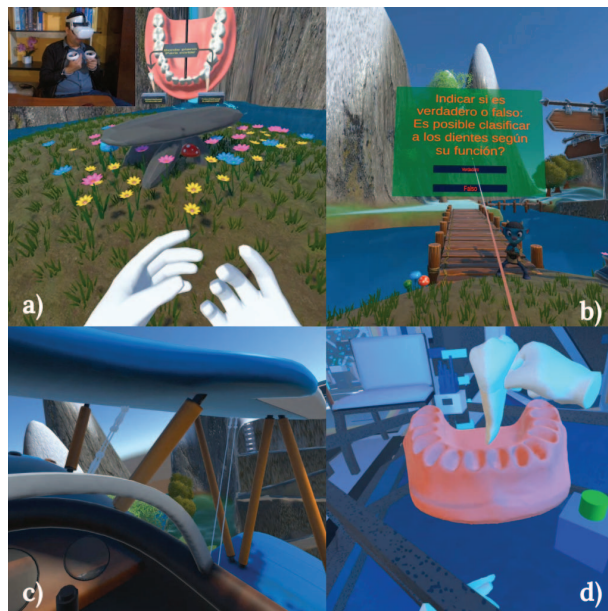


Figure 1: Main Dentoraculus Scenes: (a) Learning theory on the island of knowledge, (b) Questionnaire (c) Travel by plane as a playful activity, (d) Practice in the dental laboratory.

objectives, characters, environments and storyline, as well as a section about 3D modeling of dental pieces.

2.1 Virtual Environment Features

The application was created using Unity¹ as a development platform, which contains multiple tools that allow combining 3D models, animations, three-dimensional surround sounds and programmed logic to develop interactive experiences that can be 2D, 3D, Augmented Reality and Virtual Reality and compile them to run on different platforms. We chose the Oculus Quest 2 HDM (head-mounted display) as our platform because it has the highest penetration in the world and high performance at low cost, has head tracking with six degrees of freedom, 90-degree FOV field of view and it comes with its own controls that allow the visualization of hands, manipulation of virtual objects and navigation through the virtual world.

2.2 VR experience design

During the content planning, we worked together with the Dentistry teachers, and we defined the specific functional competencies that

¹<https://www.unity.com/>



Figure 2: Virtual Reality Development Methodology

the student should acquire through the practice of each Virtual Reality module, identifying the skills required for the achievement of each of the competencies and the criteria for their evaluation. In this demonstration we focus on three competencies: Students recognize by visual inspection each of the teeth, students know each of the functions of the teeth in the chewing process and the morphological characteristics related to it, students know the division of maxillaries in quadrants and their numbering.

The story takes place in two environments: the futuristic-looking "Dental Morphology Laboratory" where practical experiments take place, and the "Island of Knowledge" where theoretical knowledge is taught.

The characters that accompany us are: "Dentobot", a friendly flying robot that is in charge of leading the laboratory practices; "The Oracle of Knowledge", a mysterious character, who is responsible for transmitting the knowledge necessary for us to develop the missions entrusted and "Dozzy", a curious character who lives on the Island of Knowledge and ensures that we have assimilated the theoretical concepts before returning to the laboratory.

The story begins in the Dental Morphology Laboratory where Dentobot presents the challenge we must solve and explains how to use the available resources: dental teleporter, push button, empty gum and teleported teeth.

When we are about to press the Push Button to start the experiment, the Oracle of Knowledge intervenes and teleports us to the Island of Knowledge, he provides us with the knowledge required to achieve the basic skills using 3D models.

Before returning to the laboratory, Dozzy asks us to answer a questionnaire regarding to the concepts we have learned. Dozzy celebrates every time we get it right and will ask the Oracle of Knowledge to repeat the lesson if we get it wrong more than twice. Successful completion of the challenge will result in a celebratory dance, after which Dozzy will lend us his plane to facilitate our return to the lab. After a short plane ride, we will be back at the lab and start the experiment. We press the Push Button and the Dental Teleporter brings us our first tooth, which we try to place in the maxillary. In case we do not succeed in placing the tooth in the correct position, the maxillary gets annoyed and makes it explode in our hand, otherwise, it celebrates. Depending on the type of error, Dentobot will give us specific feedback.

2.3 3D modeling of teeth and maxillaries

The 3D models of maxillaries (with and without teeth) and dental pieces were elaborated using photogrammetry technique, by taking multiple photographs of the original object from different angles,

taking care that there is an overlap between each photo of approximately 80 percent, so that each point of the object appears in four or more different photos. We used the specialized photogrammetric restitution software Agisoft Metashape², which generates 3D object from the analysis of the relative positions of points common to several images. Subsequently, the 3D models resulting from the photogrammetry are decimated in Blender³ to reduce the number of polygons, then retouched and painted using the same software.

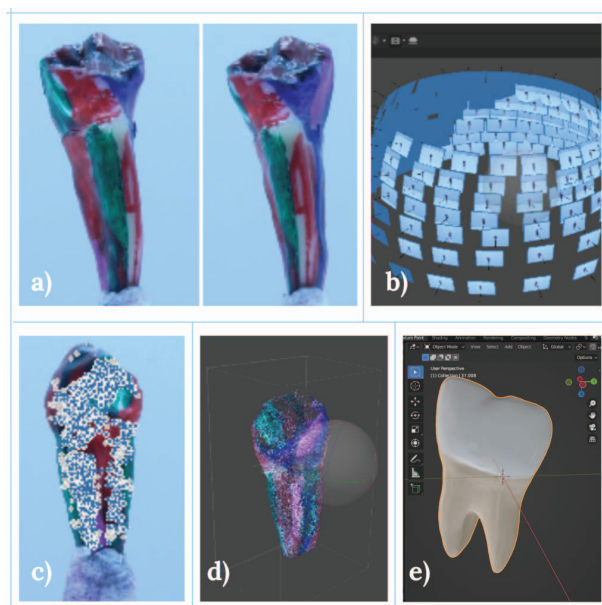


Figure 3: 3D modeling of dental pieces. a) 2D photos taken at different angles of left first premolar, b) Agisoft Metashape screenshot showing the alignment of the cameras, c) Key Points identified by the photogrammetry software, d) Construction of the sparse cloud of points, e) Incorporation of textures and painting of teeth in Blender.

3 CONCLUSIONS AND FUTURE WORKS

In this work we present an educational tool for learning dental morphology, which was tested by the professors and students of the course, obtaining very good acceptance and qualifying it as a promising tool for the training of more subjects of the dental career.

For the next version of this project we will develop a dental laboratory for the training of clinical skills in dental surgery and treatment of dental caries, with the objective of providing the students with a controlled space where they can repeatedly practice the knowledge imparted by their professors.

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²<https://www.agisoft.com/>

³<https://www.blender.org/>